Curriculum Vitae

Rajes Ghosh
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Personal Information

- Name : Rajes Ghosh
- Date of Birth : 08-11-1996 (dd-mm-yyyy)
- Nationality : Indian
- Gender : Male
- Marital Status : Single
- Languages : Bengali (mother tongue), English, Hindi
- Mobile No. : +91-9732285003, +1-4109775110

Present Position

Fulbright-Nehru Postdoctoral Fellow (Jan'26—January'28),
Department of Physics and Astronomy,
Johns Hopkins University,
Baltimore, MD 21218, USA

Academic Qualifications

- **Postdoc (2024-2026)**

  Department of Astrophysical
  Relativity,
  International Centre for Theoretical Sciences,
  Bengaluru, Karnataka

- **PhD (2019-24)** *in Physics*

  Department of Physics
  Indian Institute of Technology
  Gandhinagar, Gujarat
  CPI: 9.54

- **M.Sc (2017-19)** *in Physics*

  Department of Physics
  Indian Institute of Technology
  Gandhinagar, Gujarat
  CPI: 9.89

- **B.Sc (2014-17)** *in Physics*

  Burdwan Raj College,
  Burdwan University, West Bengal
  Marks obtained: 72%

- **12th (2012-14)** *in Bengali, English, Chemistry, Mathematics, Physics, Biology, Environmental Science*

  Arambagh High School, West Bengal
  Marks obtained: 88.2% (A+)

- **10th (2012)** *in Bengali, English, Mathematics, Physical Science, Life Science, History, Geography*

  Bachhanari Bishuddhananda High School, West Bengal
  Marks obtained: 92.3% (AA)

Achievements

- **Postdoc:** Recipient of the **Fulbright-Nehru postdoctoral fellowship,** 2-year (2026-28) position at Johns Hopkins University, USA with Prof. Emanuele Berti

- **PhD:** 1. Research showcased in **PMRF website** as Commendable Research
  2. Selected for the prestigious **Prime Minister Research Fellowship, Govt. of India**
  3. Published three papers as Letters and one paper as Rapid Communication in Physical Review D (more details below in Publications)

- **M.Sc**: 1. Winner of **President's, Director's and Institute gold medal**
  2. Winner of the **Award for outstanding research**
  3. Recipient of **Inspire fellowship, Govt. of India**

- **B.Sc:** Recipient of **Inspire fellowship, Govt. of India**

- **12th:** Recipient of **Swami Vivekananda Merit-cum-means scholarship**

Current/Future research Interests

Gravity in strong field regime, Gravitational wave signatures of modified gravity, Extreme-mass-ratio inspiral and supermassive black holes, Stability of horizonless compact objects, Black hole shadow and quasi-normal modes, Hairy black holes, Tidal response of black holes and associated Love numbers, Quantum aspects of gravity.

Research Experience

- **PhD Supervisor:** Prof. Sudipta Sarkar, IIT Gandhinagar, India
- **PhD Thesis:** "Theoretical and Observational Constraints on Theories Beyond General Relativity"; arXiv: 2406.16324 [gr-qc].
- **Journal Reviewer:** Physical Review D, General Relativity and Gravitation, Classical and Quantum Gravity, International Journal of Theoretical Physics, Modern Physics Letters A, Scientific Reports, Journal of Astrophysics & Astronomy, Physical Review A.

Computation Skills

- **Python:** Used packages such as *GWFish* for Fisher forecasting, *pyRing* for ringdown waveform analysis, and *PyCBC* for gravitational wave data analysis.
- **Mathematica:** Used packages such as *GRTensor, xTensor, xAct,* and *xPert* for tensor algebra, *QNMSpectral* and *Black Hole Perturbation Toolkit* for black hole perturbation and quasi-normal modes computation.

Publication Timeline [Inspire]

Short summary of postdoctoral research: In recent postdoctoral research, I have pursued a broad set of problems at the interface of GW physics and strong-field gravity. In particular, our work has lead to a deeper understanding of dynamical tidal Love numbers (TLNs) of black holes (BHs). We have also explored the consequences of broken spacetime symmetries (e.g., non-circularity) both for compact binaries (using tools like pyRing) and for extreme mass-ratio inspirals (EMRIs). In parallel, we are systematically developing a complementary framework to traditional post-Newtonian approaches for beyond-GR parameter estimation, based on generalized perturbed Keplerian dynamics. More recently, we have extended GW lensing beyond the weak-field analysis to incorporate strong-field effects induced by a Kerr lens, where our analysis using

GW150914-like events already demonstrates strong potential for future high-precision observations. As part of the Fulbright program at Johns Hopkins University, my present research focuses on environmental effects, specifically modeling the impact of dark matter halos around BHs on GW emission from both comparable-mass and EMRIs. This line of work is especially timely with broad relevance to fundamental physics, astrophysics, and cosmology.

Postdoc, JHU, USA (Jan, '26 - Present)

Arxiv Preprint:

1. "From Density to Mass: A New Parametrized Framework for Dark Matter Environment Around Black Holes"; R. Ghosh, N. Speeney, A. Chowdhury, C. Singha, M. A. S. Pinto, E. Berti; arXiv: 2609.35771 [gr-qc] (under review).

2. "How Universal Is the Black Hole Zeroth Law?"; R. Ghosh, S. Sarkar; arXiv: 2609.24648 [gr-qc] (under review).

3. "Accurate Black Hole Quasinormal Modes, Regge Poles, and Greybody Factors from Light Ring Ladder Symmetry"; R. Ghosh, D. Pereñiguez, J. R. Yuste, E. Berti; arXiv: 2609.25228 [gr-qc] (under review).

4. "Universal Ladder Structure Across Scales: From Quantum to Black Hole Physics"; R. Ghosh, R. P. Bhatt, S. Chakraborty, S. Bose; arXiv: 2604.06249 [gr-qc] (under review).

Postdoc, ICTS-TIFR, India (Aug, '24 - January, '26)

Journal Articles:

5. "Strong-field Gravitational Wave Lensing in the Kerr Background"; M. V. S. Saketh, R. Ghosh, A. Mishra; arXiv: 2511.23110 [gr-qc]; Phys. Rev. D 113 (2026) 8, 084056.

6. "Generalized Perturbed Kepler Problem: Gravitational Wave Imprints from Eccentric Compact Binaries"; R. Ghosh, Prasad R., K. Chakravarti, P. Kumar; arXiv: 2508.06245 [gr-qc]; Phys. Rev. D 113 (2026) 8, 084018.

7. "Can Rotating Black Holes Have Short Hairs?"; R. Ghosh, C. Singha; arXiv: 2501.08767 [gr-qc]; Phys. Rev. D 111 (2025) 4, 044008.

8. "Probing Spacetime Symmetries Using Gravitational Wave Ringdown"; R. Ghosh, A. K. Mishra, S. Sarkar; arXiv: 2412.08942 [gr-qc]; Phys. Rev. D 112 (2025) 6, 064092.

9. "Birkhoff's Theorem and Uniqueness: A Peek Beyond General Relativity"; R. Ghosh, A. K. Mishra, A. Chowdhury; arXiv: 2411.09193 [gr-qc]; Phys. Lett. B 871 (2025) 139974.

10. "Restoring Causality in Higher Curvature Gravity"; J. Edelstein, R. Ghosh, A. Laddha, S. Sarkar; arXiv: 2409.16935 [hep-th]; JCAP 09 (2026) 094.

11. "Parameterized Non-circular Deviation from the Kerr Paradigm and Its Observational Signatures: Extreme Mass Ratio Inspirals and Lense-Thirring Effect"; R. Ghosh, K. Chakravarti; arXiv: 2406.02454 [gr-qc]; JCAP 04 (2025) 037.

Short summary of doctoral research: During my PhD, I explored beyond-GR physics through three complementary directions: (i) dynamical modifications involving higher-curvature terms and additional fields, (ii) phenomenological deviations from the Kerr paradigm, and (iii) kinematical alterations of spacetime symmetries. On the dynamical side, we established the 0th law of BH thermodynamics for the full Lovelock class and developed causality-based criteria to identify consistent classical theories, while also placing constraints on the Einstein–Gauss–Bonnet coupling using GW150914 data. On the phenomenological side, we analyzed GW signatures of area-quantized BHs and developed unified numerical tools to compute quasi-normal modes (QNMs) for BHs slightly deviated from Kerr. On the kinematical side, we derived theory-agnostic results based solely on spacetime symmetries, including conditions for the existence of light rings around compact objects and constraints that rule out BH short hairs.

Ph.D., IIT Gandhinagar, India (Aug, '19 - Jul, '24)

Journal Articles:

12. "Exploring ladder symmetry and Love numbers for static and rotating black holes"; C. Sharma, R. Ghosh, S. Sarkar; arXiv: 2401.00703 [gr-qc]; Phys. Rev. D 109 (2024) 4, L041505 **(Letter)**.

13. "Formation and Stability of Area Quantized Black Holes"; K. Chakravarti, R. Ghosh, S. Sarkar; arXiv: 2310.18022 [gr-qc]; Phys. Rev. D 109 (2024) 4, 046001.

14. "Hairy Black Holes: Non-existence of Short Hairs and Bound on Light Ring Size"; R. Ghosh, Selim Sk, S. Sarkar; arXiv: 2306.14193 [gr-qc]; Phys. Rev. D 108 (2023) 4, L041501 **(Letter)**.

15. "Does the speed of gravitational waves depend on the source velocity?"; R. Ghosh, S. Nair, L. Pathak, S. Sarkar, A. Sengupta; arXiv: 2304.14820 [gr-qc]; Phys. Rev. D 108 (2023) 12, 124017.

16. "Quasi-normal modes of non-separable perturbation equations: the scalar non-Kerr case"; R. Ghosh, N. Franchini, S. Volkel, E. Barausse; arXiv: 2303.00088 [gr-qc]; Phys. Rev. D 108 (2023) 2, 024038.

17. "Regularized Stable Kerr Black Hole: Cosmic Censorships, Shadow and Quasi-Normal Modes"; R. Ghosh, M. Rahman, A. K. Mishra; arXiv: 2209.12291 [gr-qc]; Eur. Phys. J. C 83 (2023) 1, 91.

18. "Constraining the Topological Gauss-Bonnet Coupling from GW150914"; K. Chakravarti, R. Ghosh, S. Sarkar; arXiv: 2201.08700 [gr-qc]; Phys. Rev. D 106 (2022) 4, L041503 **(Letter)**.

19. "Signature of Non-Uniform Area Quantization on Black Hole Echoes"; K. Chakravarti, R. Ghosh, S. Sarkar; arXiv: 2112.10109 [gr-qc]; Phys. Rev. D 105 (2022) 4, 044046.

20. "Signature of Non-Uniform Area Quantization on Gravitational Waves"; K. Chakravarti, R. Ghosh, S. Sarkar; arXiv: 2108.02444 [gr-qc]; Phys. Rev. D 104 (2021) 8, 084049.

21. "Light Rings of Stationary Spacetimes"; R. Ghosh, S. Sarkar; arXiv: 2107.07370 [gr-qc]; Phys. Rev. D 104 (2021) 4, 044019.

22. “Causality Constraints in Quadratic Gravity”; J. D. Edelstein, R. Ghosh, A. Laddha, S. Sarkar; arXiv: 2107.07424 [hep-th]; J. High Energ. Phys. 2021, 150 (2021).

23. “Overcharging Extremal Black Holes”; R. Ghosh, A. K. Mishra, S. Sarkar; arXiv: 2106.10667 [gr-qc]; Phys. Rev. D 104 (2021) 10, 104043.

24. “Black Hole Zeroth Law in Higher Curvature Gravity”; R. Ghosh, S. Sarkar; arXiv: 2009.01543 [gr-qc]; Phys. Rev. D 102 (2020) 10, 101503 **(Rapid Communication)**.

- **M.Sc., IIT Gandhinagar, India (Aug, ’17- Jul, ’19)**

Journal Article:

25. “Overcharging higher curvature black holes”; R. Ghosh, C. Fairoos, S. Sarkar; arXiv: 1906.08016 [gr-qc]; Phys. Rev. D 100 (2019) 12, 124019.

Schools/Workshops/Conferences Attended

1. **Upcoming: Theory of tidal deformation: From nuclear theory to gravitational wave observations** discussion meeting at ICTS, India, Date: Nov 23-Nov 27, 2026.
2. **Future of GW Astronomy** discussion meeting at ICTS, India, Date: Oct 27-Oct 31, 2025.
3. **GW10@IITGN** at IIT Gandhinagar, Gujarat, India. Date: Oct 6 - Oct 10, 2025.
4. **22nd Lomonosov Conference** at Moscow State University, Russia. Date: Aug 21 - Aug 27, 2025.
5. **ICTS Summer School on Gravitational Wave Astronomy** at ICTS, India, Date: Jul 07 - Jul 18, 2025.
6. **Beyond the horizon: Testing black hole paradigm** at ICTS, India, Date: Mar 24 - Apr 04, 2025.
7. 33rd Indian Association for General Relativity and Gravitation **(IAGRG 2025)** at BITS Pilani, Rajasthan, India. Theme: Ripples in Action, Date: Jan 02 - Jan 04, 2025.
8. **IAGRG School on Gravitation and Cosmology** at ICTS, India, Date: Oct 09 - Oct 20, 2023.
9. **ICTS Summer School on Gravitational-Wave Astronomy** at ICTS, India, Date: July 24 - Aug 4, 2023.
10. **PMRF Annual Symposium** at Indian Institute of Technology Madras, India, Date: Feb 17 - Feb 18, 2023.
11. Second Chennai Symposium on Gravitation and Cosmology **(CSGC 2022)** Indian Institute of Technology Madras, India. Date: Feb 2 - Feb 5, 2022.
12. **School on Black Holes and Gravitational Waves** at Indian Institute of Technology

Gandhinagar, India. Date: Jan 17 - Jan 22, 2022.

13. **Regular Black Holes** in Quantum Gravity and Beyond - from Theory to Shadow Observations, Date: Oct 18 - Oct 22, 2021.

14. Black Holes Inside and Out **(BHIO 2021)**, Date: Sept 27 - Oct 1, 2021.

15. 31st Indian Association for General Relativity and Gravitation **(IAGRG 2020)** at IISER Kolkata, India. Theme: Quasi-normal Modes from Black Holes, Date: Dec 19 - Dec 20, 2020.

Seminars/Posters

1. **Upcoming Invited talk: Theory of tidal deformation: From nuclear theory to gravitational wave observations** discussion meeting at ICTS, India, Date: Nov 23-Nov 27, 2026.

2. **Invited CAS seminar** at JHU, USA on "Gravitational waves as probe of new physics", Date: Mar 2, 2026.

3. **Future of GW Astronomy** discussion meeting at ICTS, India. Observer for the "Probing gravity using GW observations" session (prepared end summary). Also, presented a poster (sparkler talk) titled: "Generalized Perturbed Kepler Problem", Date: Oct 30, 2025.

4. **GW10@IITGN** at IIT Gandhinagar, Gujarat. Title: "Gravitational waves as probe of new physics: Insights from inspiral dynamics to ringdown spectra", Date: Oct 07, 20225.

5. **Plenary talk: 22nd Lomonosov Conference** at Moscow State University, Russia. Title: "Universal constraints on black hole hairs: From no-short-hair theorems to observational signatures", Date: Aug 26, 2025.

6. **V. V. Narlikar Best Thesis Award Seminar:** 33rd Indian Association for General Relativity and Gravitation **(IAGRG 2025)** at BITS Pilani, Rajasthan, India. Title: "Theoretical and Observational Constraints on Theories Beyond General Relativity", Date: Jan 03, 2025.

7. 33rd Indian Association for General Relativity and Gravitation **(IAGRG 2025)** at BITS Pilani, Rajasthan, India. Title: "Are Ghosts Scary Enough?", Contributory parallel session: Poster, Date: Jan 02, 2025.

8. Astrophysics-Relativity cross-department talk on **this paper** at ICTS Bengaluru, India; Title: Higher-derivative Corrections to Kerr QNMs, Date: Nov 20, 2024.

9. Joint astrophysics-relativity seminar at ICTS, India; Title: **"Investigating Physics Beyond General Relativity and Kerr Paradigm"**, Date: Sep 12, 2024.

10. Seminar at IACS Kolkata, India; Title: "Beyond the Boundaries of General Relativity: Some Aspects of Modified Gravity", Date: Mar 1, 2024.

11. Presented our work on the test of 2nd postulate of relativity using GWs in LIGO TGR call and LIGO-India teleconference.

12. **PMRF Annual Symposium** at Indian Institute of Technology Madras, India, Poster Presentation, Date: Feb 18, 2023.

13. 32nd Indian Association for General Relativity and Gravitation **(IAGRG 2022)** meeting at IISER Kolkata, India. Title: "Constraining the topological Gauss-Bonnet coupling from GW150914", Contributory parallel session: Oral Presentation, Date: Dec 21, 2022.

14. National Physicists Meet at Indian Institute of Technology Gandhinagar, India. Title: Rings of Fire in the Sky, Date: Dec 12, 2022.

15. **Seminar** at SISSA, Italy; Title: "Some aspects of modified gravity: Causality and BH area-quantization", Dated: Apr 20, 2022.

16. **Invited Short Talk** at 2nd Chennai Symposium on Gravitation and Cosmology **(CSGC 2022)** at Indian Institute of Technology Madras, India. Title: "Causality Constraints in Quadratic Gravity", Date: Feb 2, 2022.

17. Physics Day Seminar at Indian Institute of Technology Gandhinagar, India. Title: Rings of Fire in the Sky, Date: Dec 18, 2021.

18. 31st Indian Association for General Relativity and Gravitation **(IAGRG 2020)** Meeting, Title: "Black hole zeroth law in Lanczos-Lovelock gravity", Contributory parallel session: Oral Presentation, Date: Dec 19, 2020.

Research Visits

1. Visited Prof. Sumanta Chakraborty at IACS Kolkata, India from Feb 18, 2024 to March 3, 2024 for a research collaboration on the project "Love number of compact objects and ladder structure".

2. Visited Prof. Enrico Barausse at SISSA, Italy from Apr 4, 2022 to May 10, 2022 for a research collaboration on the project "Methods to Find Quasi-Normal Modes of Modified Black Holes".

3. Visited Prof. Alok Laddha at Chennai Mathematical Institute (CMI), India from Feb 1, 2020 to Feb 25, 2020 for a research collaboration on the project "Causality Constraints in Quadratic Gravity".

Mentoring/ Teaching Experience

1. Mentoring Ankur Renduchintala (BITS Pilani, Rajasthan, India) for his dual-degree thesis project (2025-2026). Project theme: "Black Hole Spectroscopy Beyond the Kerr Paradigm".

2. Host of Journal Club in the Astrophysics-Relativity department of ICTS-TIFR in my postdoctoral tenure (2024-'26).

3. Tutor of Prof. Chiara Caprini for a 1-weak course "Probing the early universe using GW

observations" during **ICTS Summer School on Gravitational Wave Astronomy**.

4. Tutor of Prof. Sanved Kolekar for a 1-weak course during a ICTS program "**Beyond the horizon: Testing black hole paradigm**."
5. Mentored Adarsh L. Bharadwaj (M.Sc. Physics, St. Joseph's University, Bengaluru, India) for his dissertation project (Sep, 2024 - May, 2025). Project theme: "Hairy Black holes and their observational signatures".
6. Assisted Alorika Kar (Ph.D. student, ICTS, Bengaluru, India) in a project during Oct, 2024 - Aug, 2025. Project theme: "Formation and Population Studies of Supermassive Black Holes".
7. Tutor of Prof. Thomas Sotiriou for a 1-weak course "Black holes in general relativity and modified gravity" in the **IAGRG School on Gravitation and Cosmology**.
8. Taught an M.Sc. level course on "Advanced Mathematical Physics" at Institute of Advanced Research, Gandhinagar from February, 2021 to May, 2021 as a part of PMRF TA duty.
9. Taught an M.Sc. level course on "Celestial Mechanics" at Kadi Sarva Vishwavidyalaya, Gandhinagar from September, 2021 to November, 2021 as a part of PMRF TA duty.
10. Taught an M.Sc. level course on "Aspects of Light Spectrum in Astrophysics" at Kadi Sarva Vishwavidyalaya, Gandhinagar from January, 2022 to April, 2022 as a part of PMRF TA duty.
11. Taught an M.Sc. level course on "Classical Mechanics" at Kadi Sarva Vishwavidyalaya, Gandhinagar from January, 2023 to April, 2022 as a part of PMRF TA duty.
12. Several TA duties at IIT Gandhinagar in many topics including Quantum Mechanics, Electrodynamics, and Thermodynamics.

Memberships

- Lifetime member of **The Indian Association for General Relativity and Gravitation (IAGRG)** since 2023.
- Active member of the **SWAJAN Foundation** since 2012, which provides financial assistance to disadvantaged meritorious students.

Hobbies and Interests

I am also interested in recreational mathematics and playing cricket.

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I agree to the processing of my personal data contained in this document.

Professional Reference: Prof. Sudipta Sarkar, Dept. of Physics, IIT Gandhinagar, India. Contact email: sudiptas@iitgn.ac.in.

✺ For Recommendation Letters

If needed, recommendation letters will be provided by

- **Prof. Sudipta Sarkar (PhD supervisor)**, Indian Institute of Technology (IIT) Gandhinagar, India. Contact email: sudiptas@iitgn.ac.in
- **Prof. Emanuele Berti (postdoc mentor)**, Johns Hokins University, USA. Contact email: berti@jhu.edu
- **Prof. Prayush Kumar (postdoc mentor)**, International Center for Theoretical Sciences (ICTS), Bengaluru, India. Contact email: prayush@icts.res.in
- **Prof. Sumanta Chakraborty (collaborator)**, Indian Association for the Cultivation of Sciences (IACS), India. Contact email: sumantac.physics@gmail.com
- **Prof. Alok Laddha (collaborator)**, Chennai Mathematical Institute (CMI), India. Contact email: oranjok@gmail.com